Temperatura Planck în Condiții Cuantice Date Derivare și Calcul Numeric

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1. Theoretical Premises 1.1 Quantification of the impulse in a cavity of length L

For stationary oscillations in a length cavity, the resonance condition requires a whole number of half-wavelengths:

$Ln\lambda_n = \frac{2L}{n} \Longrightarrow p_n = \frac{h}{\lambda_n} = \frac{nh}{2L} = \frac{n\pi\hbar}{L}$

For the fundamental mode: $n = 1 p_1 = \frac{\pi\hbar}{L}$

1.2 Non-relativistic energy (Newton)

$$E_n = \frac{p_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

1.3 Complete relativistic energy

$$E = \sqrt{p^2c^2 + (mc^2)^2}$$

Regim ultrarelativist (, viteze): $pc \gg mc^2 v \to cE \approx pc = \frac{\pi\hbar c}{L}$

Relativistic transition condition: , where: $\pi\hbar c/L \sim mc^2 m \sim \frac{\hbar}{Lc}$

1.4 Temperature formula by equivalence to thermal energy

Equalizing quantum energy with thermal energy:$E \sim k_B T$

$$\boxed{T \approx \frac{\hbar c}{k_B L}} \quad \text{(regim relativist)}$$

$$\boxed{T \approx \frac{\pi^2\hbar^2}{2mk_B L^2}} \quad \text{(regim non-relativist)}$$

2. Input Data 2.1 Specified Parameters

Parameter	Symbol	Value
Minimum cavity length	L	$3{,}524 \times 10^{-25}$ m
Quantum Mass	m	$\frac{1}{3}m_e \approx 3{,}036 \times 10^{-31}$ kg

2.2 Physical constants used

Constant	Symbol	Value
Reduced Planck constant	$\hbar$	$1{,}0546 \times 10^{-34}$ J·s
Constanta Planck	$h = 2\pi\hbar$	$6{,}626 \times 10^{-34}$ J·s
Speed of light in vacuum	c	$2{,}998 \times 10^{8}$ m/s
Constanta Boltzmann	k_B	$1{,}381 \times 10^{-23}$ J/K
Electron mass	m_e	$9{,}109 \times 10^{-31}$ kg
Planck Length	l_P	$1{,}616 \times 10^{-35}$ m
Standard Planck temperature	T_P	$1{,}416 \times 10^{32}$ K

3. Calculus — Relativistic Regime Applied formula:

$$T_{rel} \approx \frac{\hbar \cdot c}{k_B \cdot L}$$

Calculation not cu pas:

Numerator: $\hbar \cdot c = 1{,}0546 \times 10^{-34} \times 2{,}998 \times 10^{8} = 3{,}162 \times 10^{-26}$ J·m

Denominator: $k_B \cdot L = 1{,}381 \times 10^{-23} \times 3{,}524 \times 10^{-25} = 4{,}867 \times 10^{-48}$ J·m

Result: $\boxed{T_{rel} = \frac{3{,}162 \times 10^{-26}}{4{,}867 \times 10^{-48}} \approx 6{,}496 \times 10^{21} \text{ K}}$

Scaling verification against standard Planck temperature:

Because:$T \propto 1/L$

$$T_{rel} = T_P \times \frac{l_P}{L} = 1{,}416 \times 10^{32} \times \frac{1{,}616 \times 10^{-35}}{3{,}524 \times 10^{-25}}$$

$$= 1{,}416 \times 10^{32} \times 4{,}586 \times 10^{-11} = 6{,}494 \times 10^{21} \text{ K} \quad \checkmark$$

Excellent concordance — relative error.$< 0{,}03\%$

4. Calculus — Non-Relativistic Regime with Applied Formula:$m = \frac{1}{3}m_e$

$$T_{nr} \approx \frac{\pi^2\hbar^2}{2mk_B L^2}$$

Calculation not cu pas:

$\hbar^2$**:** $(1{,}0546 \times 10^{-34})^2 = 1{,}1122 \times 10^{-68}$ J^2·s^2

Numerator ($\pi^2 = 9{,}8696$): $\pi^2\hbar^2 = 9{,}8696 \times 1{,}1122 \times 10^{-68} = 1{,}0977 \times 10^{-67}$ J^2·s^2

L^2**:**

$$(3{,}524 \times 10^{-25})^2 = 1{,}2419 \times 10^{-49} \text{ m}^2$$

Denominator: $2mk_BL^2 = 2 \times 3{,}036 \times 10^{-31} \times 1{,}381 \times 10^{-23} \times 1{,}2419 \times 10^{-49}$
$= 2 \times 3{,}036 \times 1{,}381 \times 1{,}2419 \times 10^{-103}$
$= 2 \times 5{,}208 \times 10^{-103} = 1{,}042 \times 10^{-102}$ J·kg·m^2

Result: $\boxed{T_{nr} = \frac{1{,}0977 \times 10^{-67}}{1{,}042 \times 10^{-102}} \approx 1{,}054 \times 10^{35}\ \text{K}}$

5. Checking Mass Through Relationship $m \sim \hbar/(Lc)$

The quantum mass implies the relation in the text: $m_{teor} = \frac{\hbar}{Lc} = \frac{1{,}0546 \times 10^{-34}}{3{,}524 \times 10^{-25} \times 2{,}998 \times 10^{8}}$

$$= \frac{1{,}0546 \times 10^{-34}}{1{,}0570 \times 10^{-16}} = 9{,}977 \times 10^{-19}\ \text{kg}$$

Comparison with the imposed table: $m_{impus} = \frac{m_e}{3} = 3{,}036 \times 10^{-31}$ kg

The ratio — the two masses differ by 12 orders of magnitude, confirming that m also belong to distinct physical regimes. $m_{teor}/m_{impus} \approx 3{,}3 \times 10^{12} m = m_e/3 L = 3{,}524 \times 10^{-25}$**The relativistic formula** is the one that is physically consistent for the given length.

6. Synthesis and Comparison

Formula	Physical diet	Calculated temperature
$T \approx \hbar c/k_B L$	**Relativistic** (recommended)	$\approx 6{,}50 \times 10^{21}$ **K**
$T \approx \pi^2 \hbar^2/(2mk_B L^2), m = m_e/3$	Non-relativist	$\approx 1{,}05 \times 10^{35}$ K
Standard Planck temperature ()$L = l_P$	Reference	$1{,}416 \times 10^{32}$ K
Hagedron (coarde) temperature	Reference	$\sim 10^{30}$ K
Nuclear temperature (fusion)	Reference	$\sim 10^{10}$ K

7. Physical Interpretation 7.1 Ratio to Planck Length

$$\frac{L}{l_P} = \frac{3{,}524 \times 10^{-25}}{1{,}616 \times 10^{-35}} \approx 2{,}18 \times 10^{10}$$

The length m is ~ $L = 3{,}524 \times 10^{-25}$**22 billion times greater** than the Planck length, placing this scale in the accessible subnuclear domain (comparable to the radius of a composite quark or the compactification scale in some extra-sized models).

7.2 Meaning of K Temperature$T \approx 6{,}5 \times 10^{21}$

1. **Exceeds** the nuclear fusion temperature (K) by 11 orders of magnitude10^{10}
2. **Below** the standard Planck temperature (K) by 10 orders of magnitude$1{,}416 \times 10^{32}$
3. It corresponds to a regime in which , i.e. the thermal photons have the wavelength of the order — condition of $k_BT \approx \hbar c/LL$**quantum thermal resonance** of the cavity

7.3 Quantum mass $m \approx m_e/3$

A mass of evokes the mass of the constituent up/down quartz of the proton in the current quark model (quark mass up MeV/c², quark mass down MeV/c², relative to MeV). The physical interpretation of this mass choice in the context of TauTime will be developed in the main paper.$m_e/3 \approx 2{,}2 \approx 4{,}7 m_e c^2 \approx 0{,}511$

7.4 Conclusion

The Planck temperature corresponding to the minimum length m, in relativistic regime consistent with the derived formula, is: $L = 3{,}524 \times 10^{-25}$ $\boxed{T \approx 6{,}5 \times 10^{21}\ \text{K}}$

This represents the thermal equilibrium temperature of a quantum oscillator confined to the scale , in the regime in which photon energy dominates over rest energy .$L\hbar c/Lmc^2$

8. References

1. Planck, M. (1900). Über eine Verbesserung der Wienschen Spektralgleichung. *Proceedings of the German Physical Society*, 2, 202–204.
2. de Broglie, L. (1924). *Research on the theory of quanta*. Thesis, University of Paris.
3. Shockley, W., Read, W. T. (1952). Statistics of the recombinations of holes and electrons. *Physical Review*, 87(5), 835–842.
4. Weinberg, S. (1972). *Gravitation and Cosmology*. Wiley, New York.
5. Popescu, V. (2026). Onduscule Theory (TauTime) — Temporal probability field and observational predictions. Doctoral thesis in preparation.$P(r,t)$

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